

$$3. \lim_{x \rightarrow 0} \cos(x^3 + 3x) = \underline{1}$$

$$4. \lim_{x \rightarrow 3} \frac{x^2 - 9}{x^2 + 2x - 3} = \frac{0}{12} = \underline{0}$$

$$5. \lim_{x \rightarrow -3} \frac{x^2 - 9}{x^2 + 2x - 3} \Rightarrow \frac{(x+3)(x-3)}{(x+3)(x-1)} \Rightarrow \frac{-6}{-4} \Rightarrow \underline{\frac{3}{2}}$$

$$10. \lim_{v \rightarrow 4^+} \frac{4-v}{|4-v|} \Rightarrow \text{PARA VALORES } > 4, \text{ A FUNÇÃO É CONSTANTE TENDO A } -1.$$

$$14. \lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 - 9}}{2x - 6} \Rightarrow \frac{-\sqrt{x^2(1 - \frac{9}{x^2})}}{x(2 - \frac{6}{x})} \Rightarrow \frac{-\sqrt{1 - \frac{9}{x^2}}}{2 - \frac{6}{x}} \Rightarrow \underline{-\frac{1}{2}}$$

$$20. \lim_{x \rightarrow 1} \left(\frac{1}{x-1} + \frac{1}{x^2 - 3x + 2} \right) \Rightarrow \frac{1}{x-1} + \frac{1}{(x-1)(x-2)}$$

$$\rightarrow \frac{1}{x-1} \left(1 + \frac{1}{x-2} \right) \Rightarrow \frac{1}{x-1} \left(\frac{x-1}{x-2} \right) \Rightarrow \frac{1}{x-2} \Rightarrow \underline{-\frac{1}{2}}$$

21. ASSÍNTOTA VERTICAL

(1) VER SE QUANDO $x=0$ VAI

PARA O INFINITO

$$\lim_{x \rightarrow 0} \frac{\cos^2 x}{x^2} \Rightarrow \frac{1}{0} \Rightarrow \infty, \text{ SABEMOS QUE OU } \lim_{x \rightarrow 0^+} \frac{\cos^2 x}{x^2} \text{ É } \infty, \text{ OU } \lim_{x \rightarrow 0^-} \frac{\cos^2 x}{x^2} \text{ É } \infty.$$

ENTÃO $x=0$ É ASSÍNTOTA VERTICAL

ASSÍNTOTA HORIZONTAL

$$0 \leq \frac{\cos^2 x}{x^2} \leq \frac{1}{x^2}$$

$$\lim_{x \rightarrow \infty} 0 = 0, \lim_{x \rightarrow \infty} \frac{1}{x^2} = 0$$

$$\lim_{x \rightarrow \pm\infty} \frac{\cos^2 x}{x^2} = 0$$

 $y=0$ É ASSÍNTOTA HORIZONTAL

$$22. y = \sqrt{x^2 + x + 1} - \sqrt{x^2 - x}$$

$$\frac{2x+1}{\sqrt{x^2+x+1} + \sqrt{x^2-x}}$$

ASSÍNTOTA HORIZONTAL

$$\lim_{x \rightarrow \infty} \frac{x(2 + \frac{1}{x})}{\sqrt{x^2(1 + \frac{1}{x} + \frac{1}{x^2})} + \sqrt{x^2(1 - \frac{1}{x})}} = \frac{2}{1+1} \Rightarrow \lim_{x \rightarrow \infty} f(x) = 1$$

$$\lim_{x \rightarrow -\infty} f(x) = -1$$

$$\lim_{x \rightarrow -\infty} f(x) = -1$$

ASSÍNTOTAS HORIZONTAIS: $x=1$ $x=-1$

$$23. 2x-1 \leq f(x) \leq x^2 \text{ PARA } 0 < x < 3$$

$$\lim_{x \rightarrow 1} f(x) = ?$$

$$\lim_{x \rightarrow 1} 2x-1 = 1$$

SE

$$\lim_{x \rightarrow 1} x^2 = 1$$

$$\text{ENTÃO } \lim_{x \rightarrow 1} f(x) = 1$$

$$30. g(x) = \begin{cases} 2x-x^2 \rightarrow 0 \leq x \leq 2 \\ 2-x \rightarrow 2 < x \leq 3 \\ x-4 \rightarrow 3 < x < 4 \\ \pi \rightarrow x \geq 4 \end{cases}$$

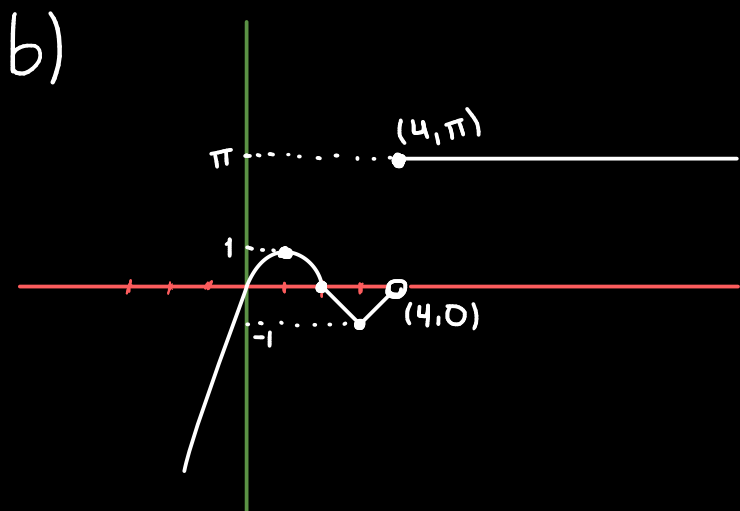
$$a) \lim_{x \rightarrow 2^-} g(x) = 2 \cdot 2 - 4 = 0$$

$$\lim_{x \rightarrow 2^+} g(x) = 2 - 2 = 0 \Rightarrow \lim_{x \rightarrow 2} g(x) = 0$$

$$\lim_{x \rightarrow 3^-} g(x) = -1 \quad \lim_{x \rightarrow 3^+} g(x) = -1$$

$$\lim_{x \rightarrow 4^-} g(x) = 0 \quad \lim_{x \rightarrow 4^+} g(x) = \pi$$

CONTÍNUA EM $(2, f(2))$ E $(3, f(3))$



$$33. \quad x^5 - x^3 + 3x - 5 = f(x)$$

$$f(1) = \cancel{1} - \cancel{1} + 3 - 5 = -2 \quad f(2) = \overset{24}{32} - \overset{+1}{8} + \cancel{6} - \cancel{5} = 25$$

$-2 < 0$ $25 > 0$

PROBLEMAS QUENTES (PÁG. 154)

$$1. \quad \lim_{x \rightarrow 1} \frac{\sqrt[3]{x} - 1}{\sqrt{x} - 1}$$

$$2. \quad \lim_{x \rightarrow 0} \frac{\sqrt{ax+b} - 2}{x} = 1$$

$$\sqrt{ax+b} - 2 = x \quad \begin{cases} ax+b = x^2 + 4x + 4 \\ \textcircled{a}x + \textcircled{b} = x(\textcircled{x+4}) + \textcircled{4} \end{cases}$$

$$\sqrt{ax+b} = x + 2$$

$$\begin{aligned} a &= x+4 \\ b &= 4 \end{aligned} \rightarrow \begin{aligned} a &= 4 \\ b &= 4 \end{aligned}$$

$$3. \quad \lim_{x \rightarrow 0} \frac{|2x-1| - |2x+1|}{x} \cdot \frac{|2x-1| + |2x+1|}{|2x-1| + |2x+1|}$$

$$\begin{aligned} &\cancel{4x^2} - \cancel{4x} - \cancel{4x^2} - \cancel{4x} - 1 \\ &\frac{|2x-1|^2 - |2x+1|^2}{x(|2x-1| + |2x+1|)} \Rightarrow \frac{-8x}{2x} = -4 \end{aligned}$$

$$9. \lim_{x \rightarrow a} [f(x) + g(x)] = 2 \quad \lim_{x \rightarrow a} [f(x) - g(x)] = 1$$

$$\begin{cases} f + g = 2 \\ f - g = 1 \end{cases} \rightarrow f \cdot g = \frac{3}{2} \cdot \frac{1}{2} = \frac{3}{4}$$

$$\underline{f = \frac{3}{2} \quad g = \frac{1}{2}}$$

$$13. f(x+y) = f(x) + f(y) + x^2y + y^2x$$

$$\text{E } \lim_{x \rightarrow 0} \frac{f(x)}{x} = 1$$

$$a) f(0) = f(0+0) \Rightarrow f(0) + f(0) \Rightarrow 2f(0)$$

SE $f(0) = 2f(0)$, $f(0)$ DEVE SER 0, DO CONTRÁRIO,

~~$\frac{f(0)}{f(0)} = 2$~~ $\Rightarrow 1 = 2$ ABSURDO, E TAMBÉM PORQUE

$$f(x+0) = f(x) + f(0) + x^2 \cdot 0 + 0^2 \cdot x$$

$$f(x) = f(x) \checkmark \text{ VERDADE}$$

$$b) \underline{f'(0) = 0}$$

$$c) f'(x) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$a = x+y$$

$$\frac{f(x) - f(x+y)}{x - x - y}$$

$$\underline{f'(x) = \frac{f(y)}{y} + x^2 + xy}$$

$$\frac{\cancel{f(x)} - \cancel{f(x)} - f(y) - x^2y - xy^2}{-y}$$

PÁGINA 165

$$3. g(x) = 4x + 7 \rightarrow g'(x) = (\underbrace{4x}_0)' + (\underbrace{7}_0)' \rightarrow \underbrace{4}_0 x + \underbrace{x}_1 \underbrace{4}_{1 \cdot 4} \rightarrow g'(x) = 4$$

$$4. g(t) = 5t + 4t^2 \rightarrow g'(t) = (5t)' + (4t^2) \Rightarrow g'(t) = 5 + \cancel{4t^2} + 4(t^2)'$$

$$g'(t) = 5 + 8t \quad \leftarrow$$

$$5. f(x) = x^{75} - x + 3 \Rightarrow f'(x) = 75x - 1$$

$$6. g(x) = \frac{7}{4}x^2 - 3x + 12 \Rightarrow g'(x) = \frac{7}{2}x - 3$$

$$8. F(t) = t^3 - e^3 \Rightarrow F'(t) = 3t^2$$

$$49. f(x) = 0.001x^5 - 0.02x^3$$

$$\hookrightarrow \frac{1}{1000}x^5 - \frac{2}{100}x^3 \rightarrow f'(x) = \frac{5}{1000}x^4 - \frac{6}{100}x^2$$

$$\rightarrow f'(x) = \frac{1}{200}x^4 - \frac{3}{50}x^2 \rightarrow f''(x) = \frac{4}{200}x^3 - \frac{6}{50}x$$

$$\rightarrow f''(x) = \frac{1}{50}x^3 - \frac{3}{25}x$$

$$62. f(x) = x^4 + 1 \quad || \quad 32x - y = 15 \Rightarrow y = 32x - 15$$

$$f'(x) = 4x^3 \rightarrow 4x^3 = 32 \int x^3 = 8$$

$$32 \cdot 2 + b = 2^4 + 1$$

$$64 + b = 17$$

$$b = \underline{-47}$$

$$\underline{32x - 47 = y}$$

$$63. f(x) = x^3 - 3x^2 + 3x - 3$$

$$\text{RETAS } || \hat{A} \quad y = 3x - 15$$

$$f'(x) = 3x^2 - 6x + 3$$

$$\hookrightarrow 3x^2 - 6x + \cancel{3} = \cancel{3}$$

$$x(3x - 6) = 0 \quad f'(0) = 3$$

$$3x - 6 = 0 \int f'(2) = 3$$

$$x = 2$$

$$3 \cdot 2 + b = 2^3 - 3 \cdot 2^2 + 3 \cdot 2 - 3$$

$$6 + b = 8 - 12 + 6 - 3$$

$$3 \cdot 0 + b = 0^3 - 3 \cdot 0^2 + 3 \cdot 0 - 3$$

$$b = -3$$

$$y = 3x - 7 \in y = 3x - 3$$

$$73. f(x) = ax^3 + bx^2 + cx + d \quad A = (-2, 6) \quad B = (2, 0)$$

$$f'(x) = (x+2)(x-2) \Rightarrow x^2 - 4 \quad \hookrightarrow f'(-2) = 0 \quad f'(2) = 0$$

$$\frac{x^3}{3} - 4x$$

$$74. f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

$$f'(1) = 4 \quad f'(-1) = -8 \quad \hookrightarrow f(2) = 15$$

$$\begin{cases} 4 = 2a + b \\ +8 = +2a - b(-1) \end{cases}$$

$$12 = 4a \quad 4 = 6 + b$$

$$a = 3 \quad b = -2$$

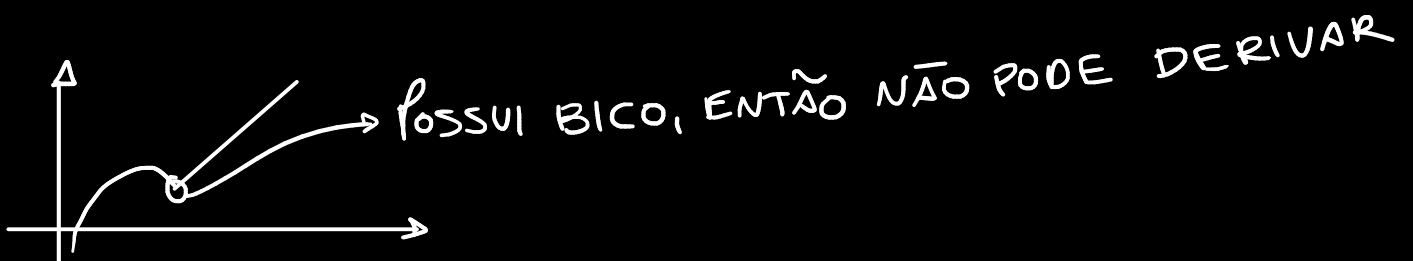
$$f'(x) = 6x - 2 \quad \int f'(x) = 3x^2 - 2x + c$$

$$15 = 12 - 4 + c$$

$$c = 7$$

$$f(x) = \underline{3x^2 - 2x + 7}$$

$$75. f(x) = \begin{cases} x^2 + 1, & x < 1 \\ x + 1, & x \geq 1 \end{cases} \rightarrow \lim_{x \rightarrow 1} f(x) = 2$$



$$76. f(x) = \begin{cases} 2x & \rightarrow x \leq 0 \\ 2x - x^2 & \rightarrow 0 < x < 2 \\ 2 - x & \rightarrow x \geq 2 \end{cases}$$

DERIVÁVEL EM $]-\infty, 0[\cup]0, 2[\cup]2, +\infty[$

$$f'(x) = \begin{cases} 2 & \rightarrow x \leq 0 \\ 2 - 2x & \rightarrow 0 < x < 2 \\ -1 & \rightarrow x \geq 2 \end{cases}$$

$$87. \lim_{x \rightarrow 1} \frac{x^{1000} - 1}{x - 1} \rightarrow a^n - b^n = (a - b) \cdot \sum_{i=0}^{n-1} a^{n-i-1} \cdot b^i$$

$$\hookrightarrow \frac{(x-1) \cdot \sum_{i=0}^{999} x^{998-i}}{(x-1)} \rightarrow \sum_{i=0}^{999} 1 \Rightarrow 999 + 1 = 1000$$

PÁGINA 179

$$1. f(x) = 3\sin x - 2\cos x \rightarrow f'(x) = 3\cos x + \underline{2\sin x}$$

$$6. g(x) = 3x + x^2 \cdot \cos x \rightarrow (x^2 \cdot \cos x)' = 2x \cdot \cos x - x^2 \sin x$$

$$\rightarrow g'(x) = -x^2 \sin x + \underline{2x \cos x} + 3$$

$$11. H(t) = \cos^2 t \rightarrow \cos t \cdot (\cos t)' + (\cos t)' \cdot \cos t$$

$$\rightarrow -\sin t \cdot \cos t - \sin t \cdot \cos t \Rightarrow -2 \cdot \underline{\sin t \cdot \cos t}$$

$$21. f(\theta) = \theta \cdot \cos \theta \cdot \sin \theta \rightarrow f'(\theta) = \theta \cdot \cos \theta \cdot (\sin \theta)' + (\theta \cdot \cos \theta)' \cdot \sin \theta$$

$$\rightarrow \theta \cos^2 \theta + (\cos \theta - \theta \sin \theta) \cdot \sin \theta$$

$$\rightarrow \theta \cos^2 \theta + \cos \theta \sin \theta - \theta \sin^2 \theta$$

$$27. f(x) = \sin x + \cos x \quad (0, \underline{1})$$

$$f'(x) = \cos x - \sin x$$

$$f'(0) = 1 \rightarrow y = x + b$$

$$\sin 0 + \cos 0 = b$$

$$b = 1$$

$$\rightarrow y = \underline{x + 1}$$

$$28. f(x) = x + \sin x \quad (\pi, \pi)$$

$$f'(x) = 1 + \cos x$$

$$f'(\pi) = 0$$

$$y = b$$

$$y = \pi$$

$$\pi + \sin \pi = b$$

$$b = \pi$$

$$31. a) g(x) = 2x \cdot \sin x \quad (\frac{\pi}{2}, \pi)$$

$$g'(x) = 2x \cdot \cos x + 2 \cdot \sin x$$

$$g'(\frac{\pi}{2}) = 2 \rightarrow y = 2x + b$$

$$\boxed{y = 2x}$$

$$2 \cdot (\frac{\pi}{2}) \cdot \sin \frac{\pi}{2} = 2 \cdot \frac{\pi}{2} + b$$

$$\pi = \pi + b$$

$$b = 0$$

32. a) $f(x) = 3x + 6\cos x$ $\left(\frac{\pi}{3}, \pi+3\right)$ $\frac{\pi}{3} = 60^\circ$

$f'(x) = 3 - 6 \cdot \sin x$ $\rightarrow y = 3(1 - \sqrt{3})x + b$

$f'\left(\frac{\pi}{3}\right) = 3 - 6 \cdot \frac{\sqrt{3}}{2} \Rightarrow 3(1 - \sqrt{3})$ $\pi+3 = 3(1 - \sqrt{3})\frac{\pi}{3} + b$

$\pi+3 = \pi - \pi\sqrt{3} + b$

$y = \underline{3(1 - \sqrt{3})x + 3 + \pi\sqrt{3}}$ $\leftarrow b = 3 + \pi\sqrt{3}$

39. $f(x) = x + 2 \cdot \sin x$

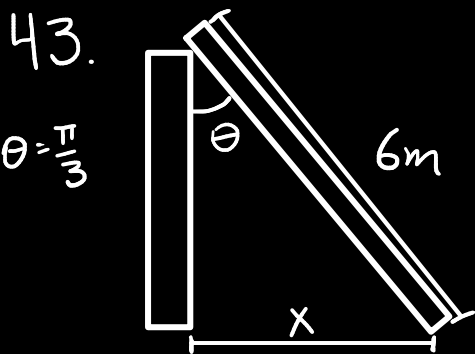
$f'(x) = 1 + 2\cos x$ $\rightarrow 1 + 2\cos x = 0$
 $\cos x = -\frac{1}{2} \rightarrow x = \frac{2\pi}{3}$
 $x = \frac{4\pi}{3}$

$f\left(\frac{2\pi}{3}\right) = \frac{2\pi}{3} + 2 \cdot \sin \frac{2\pi}{3}$

$f\left(\frac{2\pi}{3}\right) = \frac{2\pi}{3} + 2 \cdot \frac{\sqrt{3}}{2}$

$f\left(\frac{2\pi}{3}\right) = \frac{2\pi}{3} + \sqrt{3}$ $f\left(\frac{4\pi}{3}\right) = \frac{4\pi}{3} - \sqrt{3}$

$\left(\frac{2\pi}{3}, \frac{2\pi}{3} + \sqrt{3}\right), \left(\frac{4\pi}{3}, \frac{4\pi}{3} - \sqrt{3}\right)$



$(\cos \theta) = \left(\frac{x}{L}\right)' = -\sin \theta = \frac{(x)' \cdot L - L' \cdot x}{L^2}$

$-\sin \theta = \frac{L}{L^2} = -\sin \theta = \frac{1}{L}$

$f'(x) = -L \cdot \sin \theta$

45. $\lim_{x \rightarrow 0} \frac{5 \cdot \sin x}{3x} \rightarrow \frac{5}{3} \cdot \frac{\sin x}{x} \Rightarrow \frac{5}{3}$

47. $\lim_{x \rightarrow 0} \frac{\sin 3t}{\sin t} \rightarrow \frac{\sin 3t}{t} \rightarrow \frac{\sin 3t}{t} \rightarrow \sin(t+2t)$

$\rightarrow \frac{\sin t \cdot \cos 2t + \sin 2t \cos t}{t}$ $\left(\frac{\sin t}{t} \rightarrow 1\right)$

$\sin t (\cos^2 t - \sin^2 t) + (2 \sin t \cos t) \cdot \cos t$

$$\frac{\cos^2 t \cdot \sin t - \sin^3 t}{t} + \frac{2 \cdot \sin t \cdot \cos^2 t}{t}$$

$$\left(\frac{\cos^2 t \cdot \sin t}{t} - \frac{\sin^3 t}{t} \right) \rightarrow 1 + 2 \cdot 1 \cdot 1$$

$$\lim_{t \rightarrow 0} \frac{\sin 3t}{\sin t} = 3$$

$$48. \lim_{x \rightarrow 0} \frac{\sin^2 3x}{x} \Rightarrow \frac{\sin 3x}{x} \cdot \sin 3x \Rightarrow 3 \cdot 0 = 0$$

$$49. \lim_{x \rightarrow 0} \frac{\sin x - \sin x \cos x}{x^2} \Rightarrow \frac{\sin x (1 - \cos x)}{x^2}$$

$$\rightarrow \left(\frac{\sin x}{x} \right) \cdot \left(\frac{1 - \cos x}{x} \right) = 0$$

$$51. \lim_{x \rightarrow 0} \frac{\tan 2x}{x} \Rightarrow \frac{\sin 2x}{\cos 2x} \Rightarrow \frac{\sin 2x}{x \cdot \cos 2x}$$

$$\rightarrow \frac{\sin x \cos x + \sin x \cos x}{x(\cos^2 x - \sin^2 x)} \Rightarrow \frac{\frac{1}{\cos x}}{\frac{\cos^2 x - \sin x}{1}} + \frac{\frac{1}{\cos x}}{\frac{\cos^2 x - \sin x}{1}}$$

$$63. f(x) = A \cdot \sin x + B \cdot \cos x$$

$$f'' + f' = 2 \cdot f = \sin x$$

$$2(A \cdot \sin x + B \cdot \cos x)$$

$$f'(x) = A \cdot \cos x - B \cdot \sin x$$

$$A - B = 0$$

$$-B - A = 1$$

$$A = -\frac{1}{2}$$

$$A = -\frac{1}{2} = B$$

$$f''(x) = -A \cdot \sin x - B \cdot \cos x$$

$$\rightarrow A \cdot \cos x - B \sin x - A \cdot \sin x - B \cdot \cos x = \sin x$$

$$\cos x (A - B) + \sin x (-B - A) = \sin x$$

$$64. \lim_{x \rightarrow \infty} x \cdot \sin \frac{1}{x} \Rightarrow x \cdot \sin \frac{1}{x} \cdot \frac{\frac{1}{x}}{\frac{1}{x}} = \frac{\sin \frac{1}{x}}{\frac{1}{x}}$$

$$u = \frac{1}{x} \Rightarrow \lim_{u \rightarrow 0} \frac{\sin u}{u} = \underline{\underline{1}}$$

PÁGINA 246

1.